DAY 2

TOPIC

- POWER SUPPLY SYSTEM THEORY
- BASICS OF SMPS AND UPS SYSTEM
- IDENTIFY POWER CABLES AND CONNECTERS

TASKS:1

List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

RESULT

Devices that use SMPS (Switched Mode Power Supply)

1. **Desktop Computer (CPU)** – Uses an SMPS to efficiently convert AC mains power into multiple stable DC voltages (3.3 V, 5 V, 12 V) required by internal components.

2. **LED Television** – Uses an SMPS because it provides efficient, lightweight, and regulated DC power for the display and electronic circuits.
3. **Laptop Charger (Adapter)** – Uses an SMPS to convert high-voltage AC into a stable DC output while minimizing power loss and heat.

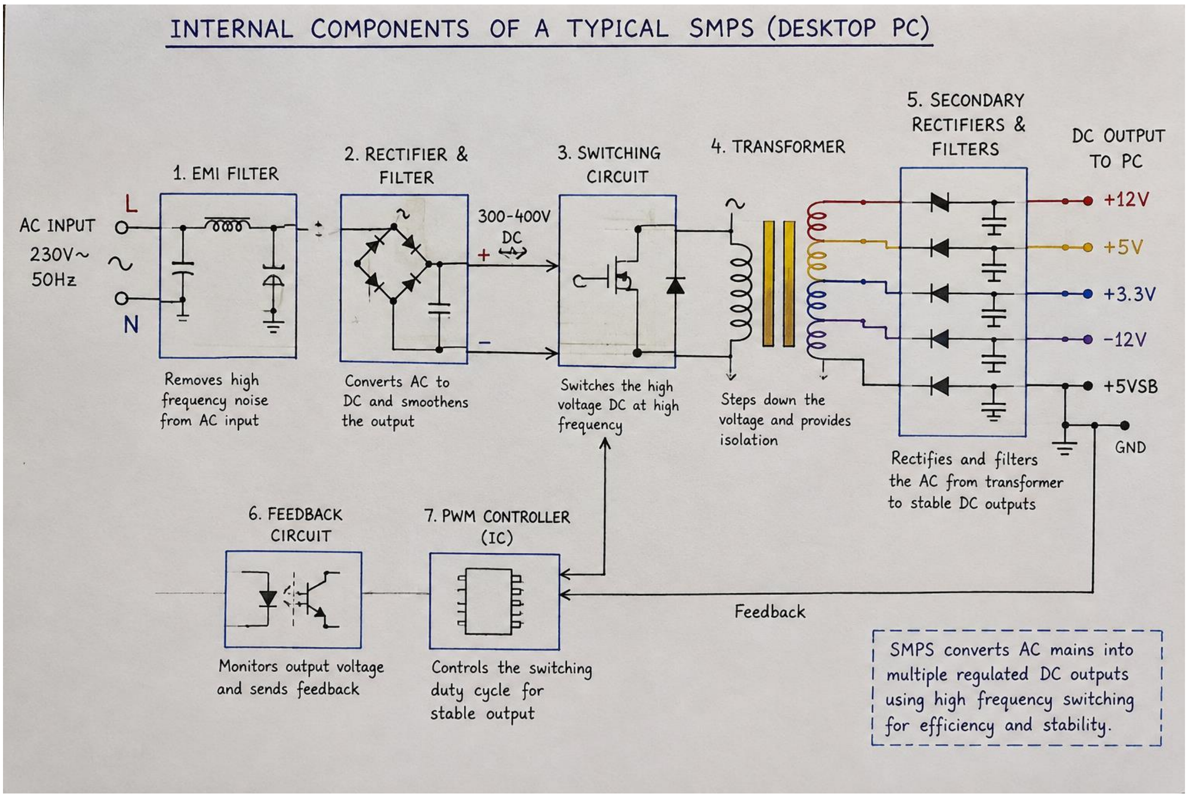
Devices that use UPS (Uninterruptible Power Supply)

1. **Desktop Computer with UPS** – Uses a UPS to provide backup power during outages, preventing sudden shutdowns and data loss.
2. **Wi-Fi Router/Modem with UPS** – Uses a UPS to keep the internet connection running during power cuts, ensuring uninterrupted communication.

TASKS:2

Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

RESULT



1. **AC Input:** Mains se 230V AC power SMPS mein enter hoti hai.
2. **EMI Filter:** Unwanted electrical noise ko filter karta hai.
3. **Rectifier & Filter:** AC voltage ko DC voltage mein convert aur smooth karta hai.
4. **Switching Circuit:** DC voltage ko high frequency par switch karta hai.
5. **Transformer:** Voltage ko required level tak reduce karta hai aur electrical isolation provide karta hai.
6. **Secondary Rectifier & Filter:** High-frequency AC ko stable DC output mein convert karta hai.
7. **DC Output:** +12V, +5V aur +3.3V jaise DC voltages PC components ko supply karta hai.
8. **Feedback Circuit:** Output voltage ko monitor karke regulation maintain karta hai.
9. **PWM Controller:** Switching ko control karke output voltage ko stable rakhta hai.

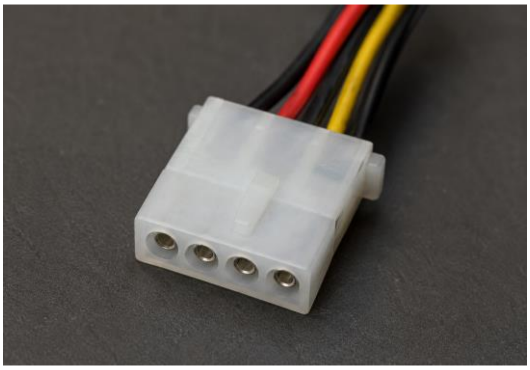
TASKS:3

Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

RESULT

ATX 24-Pin Motherboard Connector

SATA Power Connector



Molex Connector (4-Pin)

PCIe 6/8-Pin Connector

Power Connector	Devices It Powers	Safety Tip
ATX 24-Pin Motherboard Connector	Motherboard	Turn off and unplug the PC before connecting or disconnecting. Align the clip correctly and never force it.
SATA Power Connector	SATA HDDs, SSDs and optical drives	Disconnect the PSU from mains power before removing the connector. Pull the connector gently, not by the wires.
Molex Connector (4-Pin)	Older HDDs, optical drives and some case fans	Make sure the connector is aligned correctly before inserting it.
PCIe 6/8-Pin Connector	Dedicated graphics cards (GPUs)	Use only the PCIe power cable intended for the GPU and make sure the clip locks securely.

The 24-pin connector is the main motherboard power connection, SATA power is used for SATA storage/optical devices, and PCIe 6/8-pin connectors provide additional power to graphics cards

TASKS:4

Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

RESULT

Steps to Connect the SMPS (Power Supply) in a Gaming PC

1. **Turn Off and Unplug**
   - Switch off the SMPS and unplug the power cable from the wall socket.
   - Make sure all PC components are safely installed.
2. **Connect the Motherboard**
   - Take the **24-pin ATX motherboard power cable** from the SMPS.
   - Connect it to the **24-pin ATX power connector** on the motherboard.
   - Push it until the locking clip clicks into place.
3. **Connect the CPU**
   - Take the **4+4-pin EPS/CPU power cable**.
   - Connect it to the **CPU power connector (usually 8-pin)** near the CPU socket on the motherboard.
   -
4. **Connect the Graphics Card (GPU)**
   - Use the **PCIe 6-pin, 8-pin, or 6+2-pin PCIe cable**, depending on the GPU.
   - Connect it firmly to the GPU's power socket.
   - **Do not force the connector** if it does not fit.

5. **Connect SATA Storage Devices**
    - For a **SATA HDD or SATA SSD**, use the **SATA power cable** from the SMPS.
    - Connect it to the SATA power port on the drive.
6. **Connect Other Devices**
    - If an older device or accessory requires it, use the **4-pin Molex connector**.
    - Modern PCs generally use SATA power instead of Molex.
7. **Check All Connections**
    - Make sure every connector is fully inserted and its locking clip is secured.
    - Keep cables away from fans and other moving components.
8. **Power On and Test**
    - Connect the AC power cable to the SMPS and switch it on.
    - Start the PC and check that the motherboard, GPU, and storage devices are working correctly.

Quick Connection Table

Component	SMPS Cable/Connector
Motherboard	24-pin ATX
CPU	4+4-pin EPS/CPU
Graphics Card	6-pin / 8-pin / 6+2-pin PCIe
SATA HDD/SSD	SATA Power
Older accessories	4-pin Molex